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S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

Practical No. 11

Aim: Implement Linear Regression to predict continuous values using a given dataset.

Name of Student:

Roll No.:

Semester/Year: VI/III

Academic Session:

Date of Performance:

Date of Submission:

AIM: Implement Linear Regression to predict continuous values using a given dataset.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

The objectives and expected learning outcome of this practical are:

- To understand the concept of supervised learning.
- To understand Linear Regression and its importance in Machine Learning.
- To learn the relationship between dependent and independent variables.
- To implement Linear Regression using Python.
- To evaluate model performance using error metrics.
- To visualize regression results using graphs.

THEORY:

Introduction to Machine Learning

Machine Learning is a subset of Artificial Intelligence that enables systems to learn patterns from data and make predictions or decisions without explicit programming.

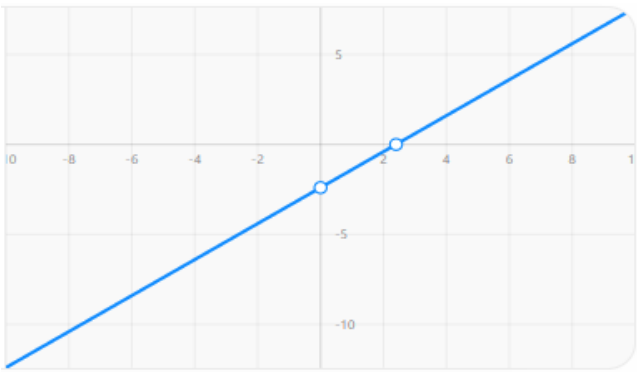
Supervised Learning

Supervised learning is a type of Machine Learning where the model is trained using labeled data. It includes:

- Regression (predict continuous values)
- Classification (predict categories)

Linear Regression

Linear Regression is one of the simplest and most widely used algorithms for predictive modeling. It establishes a linear relationship between input variables (X) and output variable (Y).



$$y = \beta_0 + \beta_1 x$$

Where:

- y = Dependent variable
- x = Independent variable
- β0 = Intercept
- β1 = Slope (coefficient)

Types of Linear Regression

1. **Simple Linear Regression** – One independent variable
2. **Multiple Linear Regression** – More than one independent variable

Working of Linear Regression

- Collect dataset
- Plot data points
- Fit a best-fit line
- Minimize error using least squares method
- Predict new values

Error Metrics

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)

Advantages

- Simple and easy to implement
- Interpretable results
- Works well with linear relationships

Limitations

- Assumes linear relationship
- Sensitive to outliers
- Cannot capture complex patterns

ALGORITHM:

1. Import required libraries
2. Load dataset
3. Split dataset into training and testing sets
4. Create Linear Regression model
5. Train the model using training data
6. Predict output using test data
7. Evaluate model using error metrics
8. Visualize results

CODE:

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

from sklearn.linear_model import LinearRegression
```

```
# Step 1: Take dataset input from user
n = int(input("Enter number of data points: "))
hours = []
marks = []
for i in range(n):
    h = float(input(f"Enter study hours for student {i+1}: "))
    m = float(input(f"Enter marks for student {i+1}: "))
    hours.append(h)
    marks.append(m)
# Step 2: Create DataFrame
df = pd.DataFrame({
    'Hours': hours,
    'Marks': marks
})
# Step 3: Define X and y
X = df[['Hours']]
y = df['Marks']
model = LinearRegression()
model.fit(X, y)
# Step 5: Predict using user input
new_hours = float(input("Enter study hours to predict marks: "))
prediction = model.predict([[new_hours]])
print("\nSlope (m):", model.coef_[0])
print("Intercept (b):", model.intercept_)
print(f"Predicted Marks for {new_hours} hours:", prediction[0])
# Step 6: Plot graph
predicted = model.predict(X)
plt.scatter(X, y, label="Actual Data")
plt.plot(X, predicted, label="Regression Line")
plt.xlabel("Study Hours")
plt.ylabel("Marks")
plt.title("Linear Regression with User Input")
plt.legend()
plt.show()
```

PERFORM THE FOLLOWING TASKS:

- Implement Linear Regression on a given dataset
- Modify dataset and observe changes
- Plot regression line
- Calculate error metrics
- Predict new values

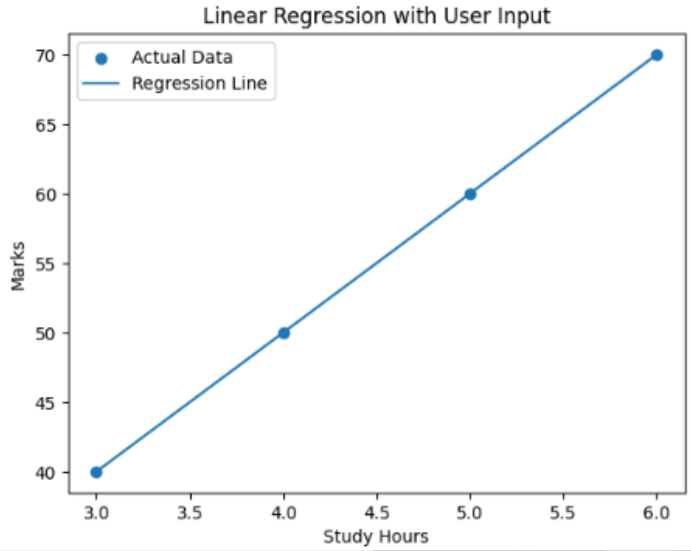
EXPECTED OUTPUT:

- Graph showing regression line
- Predicted values
- Error metrics (MSE/MAE)
- Interpretation of results

INPUT & OUTPUT:

```
Enter number of data points: 4
Enter study hours for student 1: 3
Enter marks for student 1: 40
Enter study hours for student 2: 4
Enter marks for student 2: 50
Enter study hours for student 3: 5
Enter marks for student 3: 60
Enter study hours for student 4: 6
Enter marks for student 4: 70
Enter study hours to predict marks: 7

Slope (m): 10.0
Intercept (b): 10.0
Predicted Marks for 7.0 hours: 80.0
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but LinearRegress
  warnings.warn(
```



CONCLUSION:

DISCUSSION QUESTIONS:

1. What is Linear Regression?

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2. Difference between simple and multiple regression?

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3. What is the role of slope and intercept?

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4. What are error metrics in regression?

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5. What are limitations of Linear Regression?

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REFERENCES:

- https://scikit-learn.org/stable/modules/linear_model.html
- https://www.geeksforgeeks.org/linear-regression-python-implementation/
- https://towardsdatascience.com/linear-regression-using-python